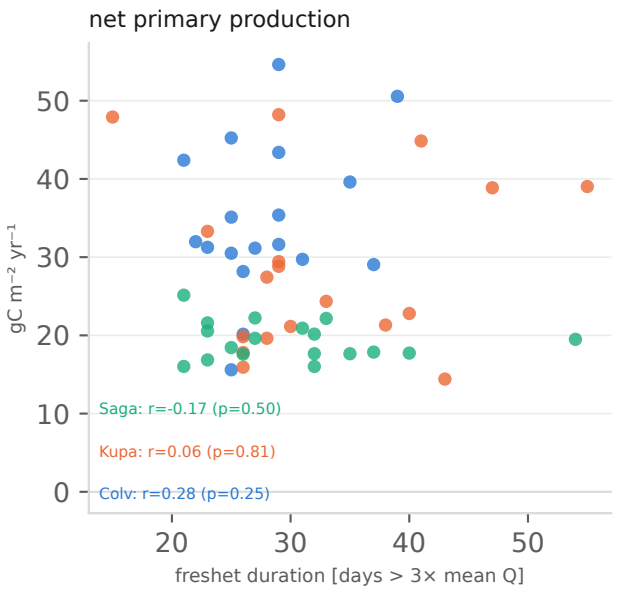
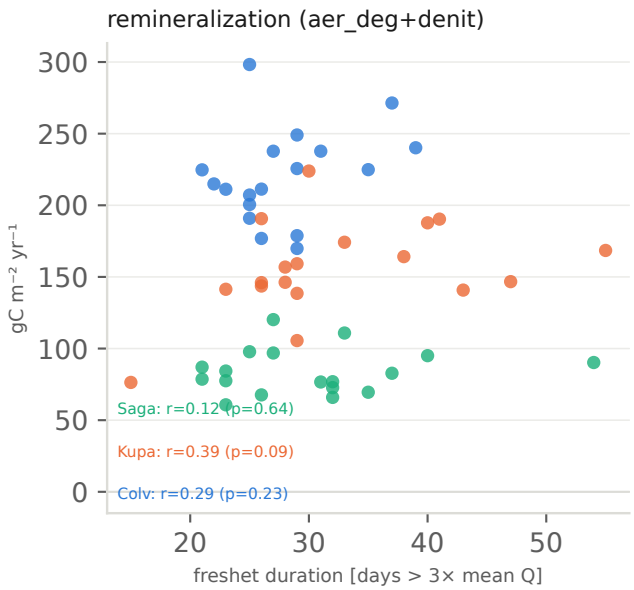
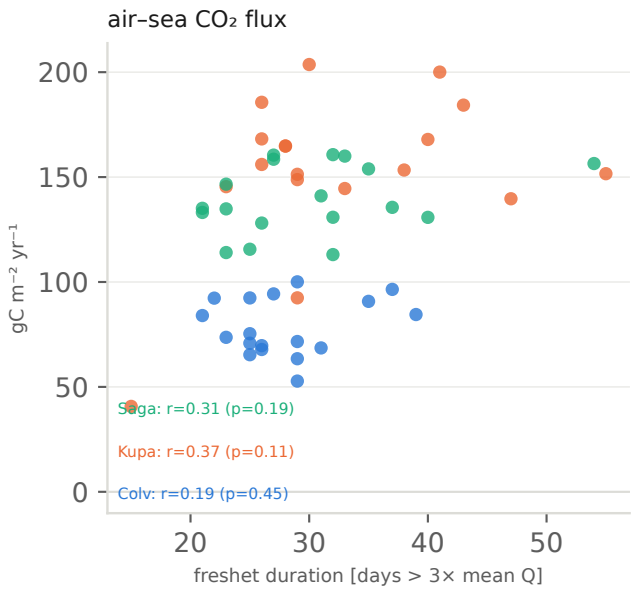
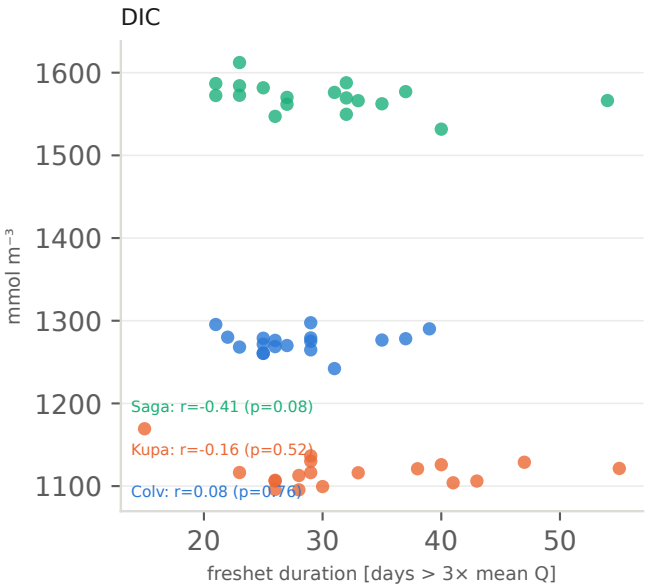
Does freshet duration/intensity shape estuarine biogeochemistry?

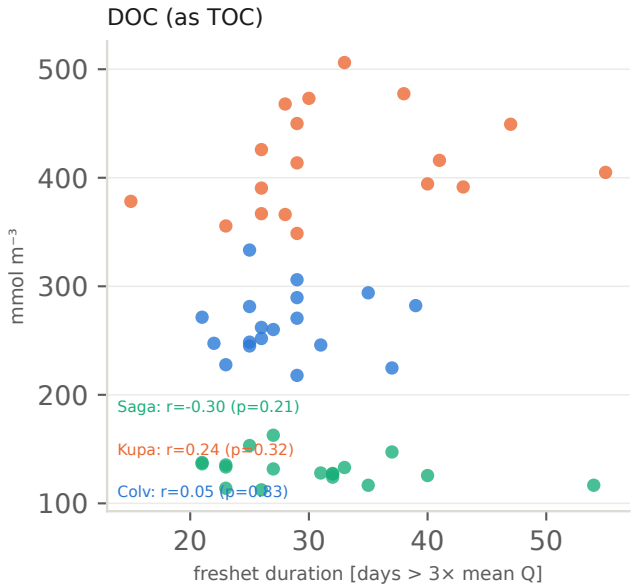
Freshet duration (days > BREAKUP_Q_FACTOR× that year's mean Q) vs. annual budgets/means of each response variable, 2006–2023, one point per site-year. Rate variables (NPP/REMIN/FCO2) are

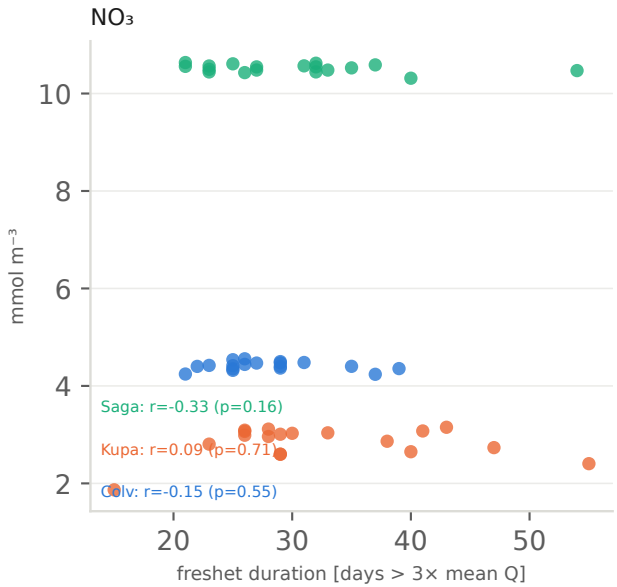












FORTE Hypothesis 1: spring-freshet timing/intensity modulate carbon/nutrient fluxes and air-sea CO2 flux. Uses runs/interannual_full/ (18 post-warmup years). Upstream (river) boundaries are CONSTANT for every solute except T and TOC -- same limitation as the original 44-yr record. Marine (mouth) boundaries, by contrast, are genuinely time-varying here for EVERY solute (S, DIC, ALK, NO3, NH4, PO4, dSi, O2, TOC -- ECCO-Darwin v5, 2005-2023), which is new relative to the original figures (only T was marine-time-varying there). Structure seen here reflects BOTH the model's own internal dynamics acting on a constant upstream source AND a genuinely time-varying ocean signal propagating in from the mouth. See CLAUDE.md -> "Interannual forcing" -> "Full-forcing interannual (2005-2023)".